AI 101 for Mechanical Engineers · Module 12

# Windows and horizons

## Lesson 12.3

# What you will learn

> Distinguish row positions from actual time, define sampling rate, window bounds and forecast horizons, and prevent temporal leakage from future or overlapping windows.

# Lesson overview

Module 12 · Lesson 3

## Windows and horizons

AI 101 for Mechanical Engineers

### 1 Use timestamps, not row positions

Every window feature needs a stated window length, alignment and sampling rate.

**RMS** summarizes signal magnitude.

**Spectral features** summarize frequency content.

**Complete 15-minute data: 96 samples per day**

$(24\ \text{h} \times 60\ \text{min/h}) / (15\ \text{min/sample}) = 96$

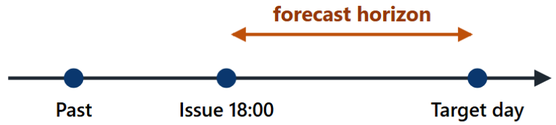
A shift of 96 intervals = 24 h; 96 consecutive timestamps span 23 h 45 min.

After filtering or missing intervals, a 96-row shift may span more than one day. **A row number is not a clock.**

Align lagged features and targets using actual timestamps.

### 2 Define the forecast horizon

The horizon is the time from forecast issue to target time.




**Next-day load forecast issued at 18:00**

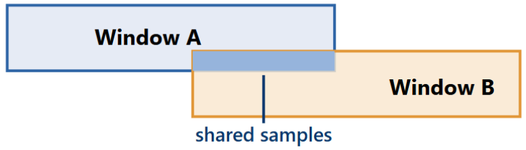
✓ **Observed load up to 18:00:** available.

! **Previously issued weather forecast:** may be available; keep its uncertainty explicit.

✗ **Tomorrow's measured weather:** exclude.

### 3 Keep overlapping windows apart

Adjacent windows can share many raw samples.




Assigning related windows randomly to training and test sets can make performance look too good.

**Use a split suited to the question:** chronological order, suitable temporal gaps, or event- or equipment-level holdouts.

Record missing intervals, sampling changes and incomplete-window rules.

**Key takeaway** **Define time and information availability before training. A row number is not a clock.**

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# Window features need documented settings

- RMS: square root of mean squared samples; same unit as signal
- Spectral features summarise frequency content [92]
- Document sampling frequency, window length and alignment

# State the spectral method and its scaling

## Power spectral density

- Squared signal units per hertz [93]

## Power spectrum

- Squared signal units [93]

# 15-minute sampling gives 96 samples per day

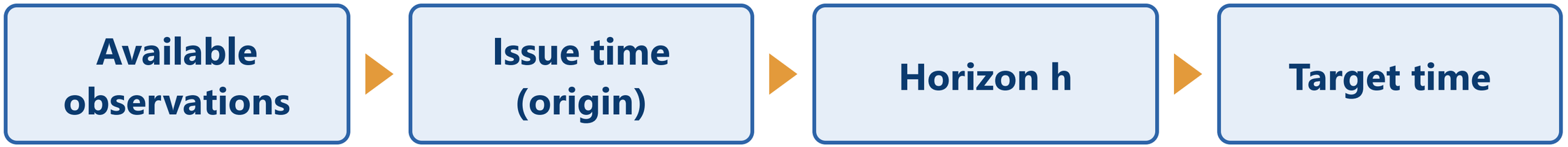
$$24\ \text{h} \times 60\ \text{min/h} \div 15\ \text{min} = 96$$

- A shift of 96 intervals moves exactly 24 hours
- 96 consecutive timestamps span 23 hours 45 minutes
- Holds only for complete, regular data

# A filtered row index is not a clock

- Removed rows or missing intervals break these relationships
- Align lagged features and targets by actual timestamps
- Record missing intervals and changes in sampling

# Draw an information-availability timeline




- h: steps or time from forecast origin to target [18]
- Use available observations and already-issued forecasts
- Exclude future readings and information generated after the decision

# Next-day load forecast issued at 18:00

| Input | Use? |
| --- | --- |
| Observed load up to 18:00 | May be available |
| Previously published weather forecast | May be available, with its uncertainty |
| Tomorrow’s measured weather | Not available; exclude |

# Overlapping windows threaten independence

## Random window split

- Adjacent windows share raw samples
- Can give optimistic performance [54]

## When independence is required

- Chronological splits
- Suitable temporal gaps
- Event- or equipment-level holdouts

No single imputation or split rule is prescribed; justify the treatment for the task.

CHECK YOUR UNDERSTANDING

# Does shifting 96 rows after deleting night-time rows still mean one day?

Answer before moving on.

CHECK YOUR UNDERSTANDING

## Does shifting 96 rows after deleting night-time rows still mean one day?

> No. After filtering, 96 rows may span several calendar days. Align by timestamps and the required time offset.

# Key takeaways

- Define sampling rate, window, alignment, issue time and horizon
- Use timestamps, not row positions, after filtering or gaps
- Exclude future readings and post-issue information
- Use temporal gaps or suitable holdouts for overlapping windows

# References

**[18]** R. J. Hyndman and G. Athanasopoulos, Forecasting: Principles and Practice, 3rd ed. Melbourne, Australia: OTexts, 2021. [Online]. Available: https://otexts.com/fpp3/

**[54]** D. R. Roberts et al., “Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure,” Ecography, vol. 40, no. 8, pp. 913–929, 2017. doi: 10.1111/ecog.02881

**[92]** R. B. Randall, Vibration-Based Condition Monitoring: Industrial, Automotive and Aerospace Applications, 2nd ed. Hoboken, NJ, USA: Wiley, 2021. doi: 10.1002/9781119477631

**[93]** SciPy developers, “scipy.signal.welch,” in SciPy v1.18.0 Manual. [Online]. Available: https://docs.scipy.org/doc/scipy/reference/generated/scipy.signal.welch.html Accessed: Oct. 20, 2026.

Numbers match the course reference list. The written lesson is the authoritative text.